Complex Numbers

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Learning Outcome

At the end of this lecture, you should be able to:

□ Understand what is a complex number and identify the real and imaginary part of a complex number.

□ Perform the algebraic operations involving complex numbers.

□ Understand and perform the algebraic laws on complex numbers.

□ Understand what is a conjugate of a complex number and its properties.

Introduction

When you first learnt how to count, you used natural number

$$\mathbb{N} = \{1, 2, 3, 4, \ldots\}.$$

Then, you found that you need something to represent the situation that you have none of the thing you're counting or to answer problems like

$$1 - 1.$$

So, you learnt about the existence of number "0".

But soon you realised you couldn't solve problems like

$$x + 2 = 1.$$

So you learnt about the negative numbers and the integers

$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}.$$

Introduction

Living happily until one day, you encounter the problems like

$$2x = 1.$$

So, you come to the rational numbers

$$\mathbb{Q} = \{\frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0\}.$$

So now, you have no problem doing $+, -, \times, \div$ until you see the problems like

$$x^2 = 2.$$

To solve this, we add in the irrational number to the set and form the set of real number $\mathbb{R}$.

So now, are you able to solve all the problems?

Introduction

How about this problem

$$x^2 = -1?$$

To solve this problem, we extend our number system again by introducing the complex number i (also known as imaginary unit), with the property

$$i^2 = -1, \text{ or } i = \sqrt{-1}.$$

$x^2 = 4$

$x = \pm\sqrt{4}$

$$x = \pm\sqrt{-b^2} = \pm\sqrt{(-1)b^2} = \pm\underbrace{(\sqrt{-1})}_{i}\underbrace{\sqrt{b^2}}_{b}$$

Remark:

$x = \pm\sqrt{-b^2}$

- The equation $x^2 = -b^2$ has the solution of $x = \pm ib$.
- Any quadratic equation $ax^2 + bx + c = 0$, $a, b, c \in \mathbb{R}$ has the solutions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Complex numbers

A complex number z is any expression of the form $z = a + bi$ where $a, b \in \mathbb{R}$ and $i = \sqrt{-1}$.

Here, a is called the real part of z, denoted by $\mathrm{Re}(z)$, while b is called the imaginary part of z, denoted by $\mathrm{Im}(z)$.

We use $\mathbb{C}$ to represent the set of all complex numbers,

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}.$$

If $b = 0$, $a + bi = a + 0i = a$ is a real number.
If $a = 0$, $a + bi = 0 + bi = bi$ is a pure imaginary number.
A complex number $a + bi$ where $b \neq 0$ is called imaginary.

Two complex numbers $a + bi$ and $c + di$ are equal if and only if $a = c$ and $b = d$.

Complex numbers

Example: If $z = 3 - 4i$, what are the real part and the imaginary part of z?

$Re(z) = 3 \qquad Im(z) = -4$

Find i^{10}, i^{21}, i^{44}, i^{75} and i^{4k} for any integer k $\qquad i^2 = -1$

$i^{10} = (i^2)^5 = (-1)^5 = -1$

$i^{21} = i^{20+1} = i^{20} \cdot i = (i^2)^{10} \cdot i = (-1)^{10} \cdot i = i$

$i^{44} = (i^2)^{22} = (-1)^{22} = 1$

$i^{75} = i^{74+1} = (i^2)^{37} \cdot i = (-1)^{37} \cdot i = -i$

$i^{4k} = (i^2)^{2k} = (-1)^{2k} = 1$

For any integer k, $i^{4k} = 1$, $i^{4k+1} = i$, $i^{4k+2} = -1$, $i^{4k+3} = -i$

Let $z = a + bi$ and $w = c + di$, where $a, b, c, d \in \mathbb{R}$.

$(2+3x) + (3+7x) = 5 + 10x$

$(2+3x)(3+7x) = 6 + 14x + 9x + 21x^2$

- **Equality:** $z = w \iff a = c$ and $b = d$
- **Addition:** $z + w = (a+bi) + (c+di) = (a+c) + (b+d)i$
- **Subtraction:** $z - w = (a+bi) - (c+di) = (a-c) + (b-d)i$
- **Multiplication:** $zw = (a+bi)(c+di) = ac + adi + bci + bdi^2$
  $= (ac - bd) + (ad + bc)i$

$a^2 - b^2 = (a+b)(a-b)$

- **Division:**
  $$\frac{z}{w} = \frac{a+bi}{c+di} = \frac{(a+bi)}{(c+di)} \cdot \frac{(c-di)}{(c-di)} = \frac{(ac+bd) + (bc-ad)i}{c^2 - (di)^2} \quad d^2i^2$$
  $$= \frac{(ac+bd) + (bc-ad)i}{c^2 + d^2}$$
  $$= \frac{ac+bd}{c^2+d^2} + \frac{bc-ad}{c^2+d^2}i$$
- **Conjugate:**
  $\bar{z} = a - bi$
  $\bar{w} = c - di$

Operations on complex number

Remark: Let $z, w, v, u \in \mathbb{C}$

- $z + w = w + z$ (Commutative Law)
- $(z + w) + v = z + (w + v)$ (Associative Law)
- $zw = wz$ (Commutative Law)
- $(zw)v = z(wv)$ (Associative Law)
- $z(w + v) = zw + zv$ (Distributive Law)
- $(z + w)v = zv + wv$ (Distributive Law)
- $\frac{z}{u} \cdot \frac{w}{v} = \frac{zw}{uv}$ if $u, v \neq 0$

Example:

1. Simplify (a) $x^2+1 = 0 \Rightarrow x = -i, i$ (b) $(x+p)^2+q^2 = 0 \Rightarrow x = -p-iq, -p+iq$

$= x^2-(-1)$

$= x^2-i^2$

$= (x+i)(x-i)$

$= (x+p)^2-(-1)q^2$

$= (x+p)^2 - i^2q^2 \quad (iq)^2$

$= (x+p+iq)(x+p-iq)$

2. Solve $x^2-2x+8=0$. $a=1$, $b=-2$, $c=8$

Quadratic formula: $x = \dfrac{-b \pm \sqrt{b^2-4ac}}{2a} = \dfrac{2 \pm \sqrt{(-2)^2-4(1)(8)}}{2(1)}$

$= \dfrac{2 \pm \sqrt{-28}}{2}$

$= \dfrac{2 \pm \sqrt{(-1)(4)(7)}}{2} = \dfrac{2 \pm i(2\sqrt{7})}{2}$

$= 1 \pm \sqrt{7}i$

3. Let $z = 2+4i$ and $w = -1+3i$.

$z+w = 1+7i \leftarrow 4i+3i$ ($2+(-1)$)

$z\bar{z} = (2+4i)(2-4i) = (2)^2-(4i)^2 = 4-(-16) = 20.$

$z-w = 3+i$

$\dfrac{z}{w} =$

$zw = (2+4i)(-1+3i) = -2+6i-4i+12i^2 = -2+2i-12 = -14+2i$

$$\frac{z}{w} = \frac{2+4i}{-1+3i} \cdot \frac{-1-3i}{-1-3i} = \frac{-2-6i-4i-12i^2}{(-1)^2-(3i)^2}$$

$$= \frac{10-10i}{1-(-9)}$$

$$= 1-i$$

Conjugate

Let $z = a + bi \in \mathbb{C}$, then its conjugate $\overline{z} = a - bi$.

- $\overline{z+w} = \overline{z} + \overline{w}$. $\Rightarrow$ $\overline{z_1 + z_2 + z_3} = \overline{z_1} + \overline{z_2 + z_3}$
- $\overline{z-w} = \overline{z} - \overline{w}$. $= \overline{z_1} + \overline{z_2} + \overline{z_3}$
- $\overline{zw} = \overline{z}\ \overline{w}$. $\Rightarrow$ $\overline{z_1 z_2 z_3} = \overline{z_1}\ \overline{z_2 z_3} = \overline{z_1}\ \overline{z_2}\ \overline{z_3}$
- $\overline{\left(\frac{z}{w}\right)} = \frac{\overline{z}}{\overline{w}}$.
- $z = \overline{z} \leftrightarrow z$ is a real number. $\overline{a+bi} = a - bi$ $\Rightarrow a = a,\ b = -b$ $\Rightarrow b = 0$
- $z\overline{z} = a^2 + b^2$.

Exercise: Try to verify these.

Remark: $\overline{z_1 + z_2 + \dots + z_n} = \overline{z_1} + \overline{z_2} + \overline{z_3} + \dots + \overline{z_n}$

$\overline{z_1 z_2 \cdots z_n} = \overline{z_1}\ \overline{z_2}\ \overline{z_3} \cdots \overline{z_n}$

Conjugate

Let $z = 2 - 7i$ and $w = 3 + i$.

$z + w =$ $5 - 6i$

$\overline{z + w} =$ $5 + 6i$

$\bar{z} + \bar{w} =$ $(2 + 7i) + (3 - i) = 5 + 6i = \overline{z+w}$

$z - w =$

$\overline{z - w} =$

$\bar{z} - \bar{w} =$

$zw =$

$\overline{zw} =$

$\bar{z}\,\bar{w} =$

$\frac{z}{w} =$

$\frac{\bar{z}}{\bar{w}} =$

Exercise.